IBRAHIM ROGERS

MECHANICAL DESIGN | TEST SYSTEMS | HARDWARE DEVELOPMENT

B.S. Mechanical Engineering, Northeastern University · Junior standing · Expected May 2028

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SELECTED WORK

01	Liquid Valve Endurance Tester & Leak-Test Fixture	Parker Hannifin · 2026
02	Contamination Root-Cause Investigations	Parker Hannifin · 2026
03	Packaging Redesign & Machine-Vision Inspection	Parker Hannifin · 2026
04	Printed Fixturing — Nest, Petri Tray & Purge Manifold	Parker Hannifin · 2026
05	Refreshable Braille Tactile Display	Forge · 2026
06	Vision-Guided Pan/Tilt Tracking System	Independent · 2025–26
07	Gamma Shielding Test System	Melagen Labs · 2024
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Liquid Valve Endurance Tester & Leak-Test Fixture

Mechanical & Test Engineering Co-op — Parker Hannifin, Precision Fluidics · Hollis, NH · May 2026 – Present

OVERVIEW

Context: Precision fluid-control valves for medical and life-science systems. My work spans test-system design, fixture design, and root-cause investigation on returned product.

My Role: Requirements definition and hydraulic architecture for a closed-loop endurance tester, through to building and commissioning the loop; mechanical design of a supplier-side leak-test fixture.

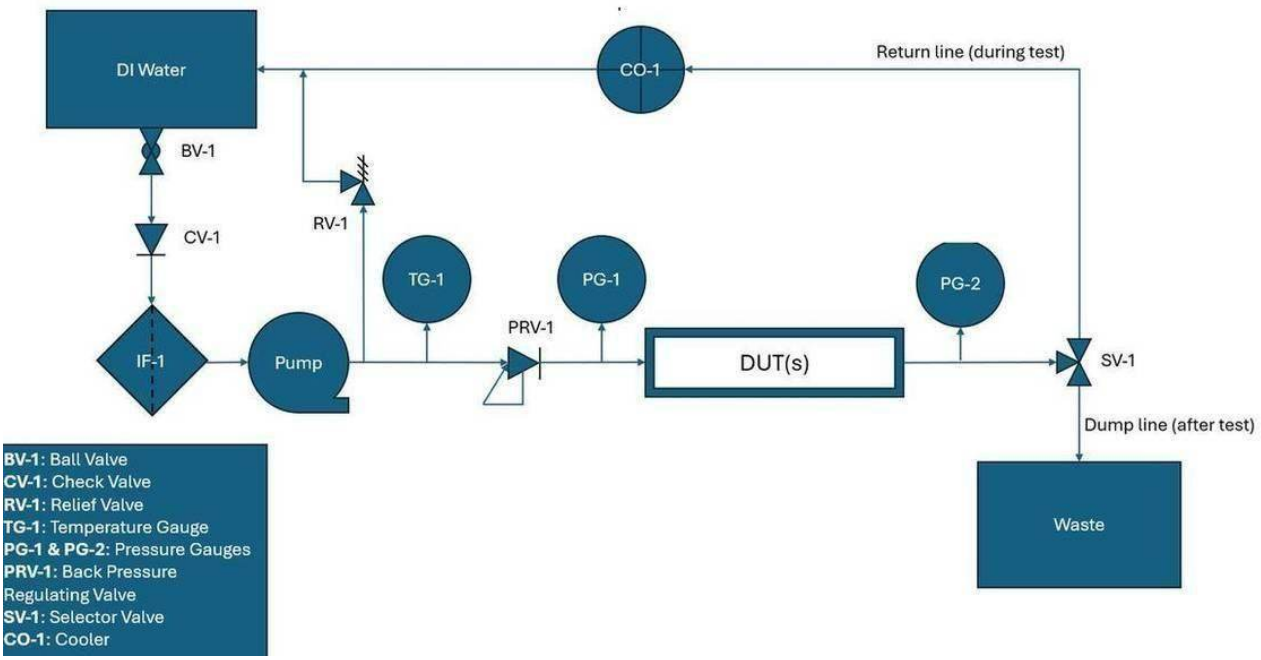
CLOSED-LOOP DI-WATER VALVE ENDURANCE TESTER

Consolidated conflicting inputs from quality, manufacturing, and engineering into a single defined test-system scope, then developed the architecture and produced build-ready drawings, schematics, and P&IDs to match.

Test capacity	**30–60 valves cycled simultaneously**
Operating pressure	**120 psig**
Fluid temperature control	**Held within ±5°F of ambient**
Working fluid	**DI water, continuous recirculation**
Architecture	**Closed loop, back-pressure regulated, chilled**
Protection	**Relief valve to tank, inline filtration, isolation valves**

Sized pump, chiller, reservoir, manifold, and instrumentation, and integrated pressure control, relief protection, and filtration for continuous recirculation. I then built the loop from those drawings: plumbed the suction and return runs, installed filtration and isolation valves, wired and commissioned the pump, and leak-checked the assembled system.

LOOP SCHEMATIC



Test loop P&ID. DI water draws from the reservoir through an isolation valve, check valve, and inline filter to the pump. A relief valve returns to tank; a back-pressure regulating valve sets test pressure upstream of the DUT bank. Gauges sit either side of the DUT so pressure drop can be read across it, and a selector valve routes the outlet either back through the cooler to the reservoir during test or to waste on drain-down.

AS-BUILT HARDWARE



Pump, filter, and valve assembly. Jet pump with accumulator, inline filter housing, and isolation valves plumbed with braided suction and return lines. Built from the schematic above.



Loop routed to reservoir. Return and suction runs from the pump cart to the bulkhead fitting on the reservoir, with the accumulator tank at left. Assembled and leak-checked before instrumentation.



Loop as tested. Reservoir and isolation valve on the upper shelf, pump on the lower shelf, with the outlet run through the gauge to the drain pan.



Configuration after the bleed fix. The added bleed point purges trapped air from the line, which is what was preventing CV-1 from reaching cracking pressure.

MVP COMMISSIONING

MVP testing ran the loop end to end on DI water. Two findings came out of the first runs: air trapped in the line, and CV-1 not reaching cracking pressure in service.

They turned out to be one finding. Entrained gas is compressible, so it absorbed the pressure rise that should have been developing across the check valve, and the loop never built enough differential pressure to crack CV-1. The symptom that looked like a valve problem was a gas problem.

Adding a bleed valve to purge the trapped air resolved it. The fix carries a design lesson worth more than the fix: bleed points belong in the P&ID from the start, placed at the high points of the routing, rather than added after commissioning finds them.

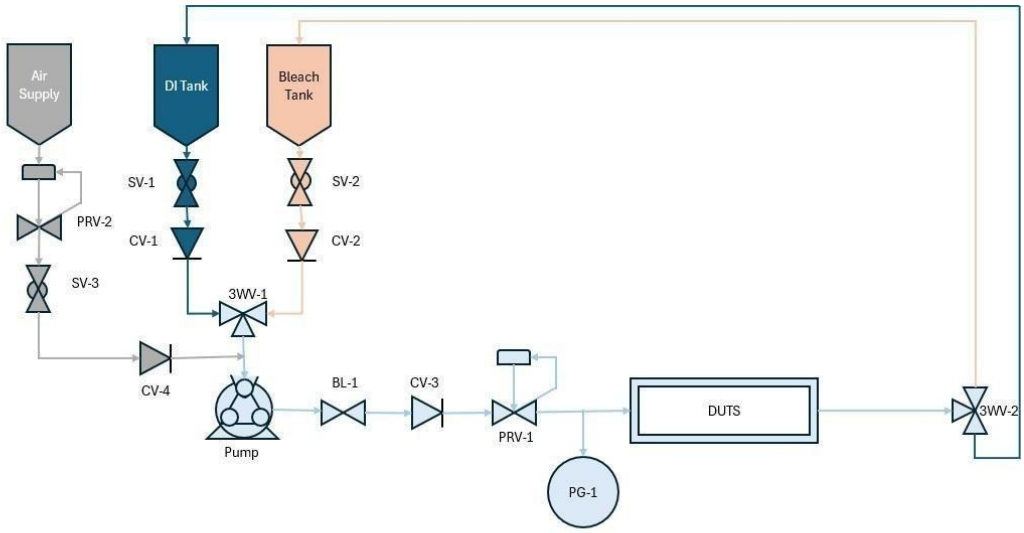
BLEACH CORROSION CYCLING

A second rig runs a multi-media corrosion study on wetted internals across **15 valves in series**. Rather than a continuous soak, the sequence cycles: dose the string with **10% sodium hypochlorite**, purge with air, run DI water through, purge again, repeat. Solenoid valves handle the media switching under electrical actuation, and the air purge between media keeps the two from mixing in the line. Wet-dry cycling is the more aggressive test, because a film left on a surface concentrates as it evaporates and has oxygen freely available to it.

Sizing the dosing pump started from a fill-time requirement rather than a flow spec: the flow needed to fill the internal volume of every valve in the series string, plus the inlet, outlet, and intervalve tubing, **inside 2 seconds**. The total internal volume of the string sets that number, not any single component in it, and it sized the dosing pump at **1,800 GPD, about 1.25 GPM**.

A peristaltic head was specified at that rating and a basic unit has been tested. The reason is containment rather than flow. In a peristaltic head the fluid only ever contacts tubing, so there is no impeller, seal, or housing to qualify against hypochlorite, and changing working fluid is a tube swap. Still to verify: the discharge pressure the head develops against the pressure drop of the series string at that flow, and tube life under repeated hypochlorite exposure.

MEDIA-SWITCHING SCHEMATIC



PRV Pressure-regulating Valve
SV Solenoid Valve (2-way)
CV Check Valve
3WV Solenoid Valve (3-way)
BL Bleed Valve
PG Pressure Gauge

Corrosion tester P&ID. Three supplies feed a common line through check valves: DI water, 10% hypochlorite, and regulated shop air. Two-way solenoids isolate each tank and a three-way solenoid selects which medium reaches the pump, so the sequence runs under electrical control rather than by hand. A back-pressure regulating valve sets test pressure upstream of the DUT string, a bleed valve sits on the pump outlet, and a second three-way valve routes the outlet back to the tanks or to drain.

PRESSURE-DECAY LEAK-TEST FIXTURE

Designed a fixture in Autodesk Inventor for supplier-side incoming inspection of a molded fluid-control component. Unlike the two test rigs above, the resolution targets here are tied to a specific product, so the numbers stay out and only the method and the reasoning behind it are shown.

Method selected	**Pressure decay**
Seal architecture	**Guarded multi-port**
Probe sizing basis	**Worst-case tolerance stack, MMC/LMC**
Part under test	**Molded component with real dimensional variation**
Application	**Supplier-side incoming inspection**

Target leak rate and trapped volume are Parker proprietary and withheld.

DESIGN REASONING

I selected pressure decay over steady-state make-up flow because flow instrumentation fine enough to resolve the required leak rate was not available at acceptable cost. Locating and sealing functions are separated so the probe seals against the orifice without over-constraining a molded part that carries real dimensional variation.

Contamination Root-Cause Investigations

Mechanical & Test Engineering Co-op — Parker Hannifin, Precision Fluidics · Hollis, NH · May 2026 – Present

OVERVIEW

Led two independent root-cause investigations into recurring return failures on precision fluid-control valves. Each involved a distinct contamination mechanism and a separate population of returned parts.

INVESTIGATION A — ORIFICE CORROSION

Scope	**Several-hundred-unit return population**
Variables isolated	**Supplier, stock, cleaning, floor process**
Outcome	**Contamination source isolated; mitigation validated**

I designed isolation experiments across each candidate variable instead of inspecting finished returns alone. That let the investigation separate a supplier-side cause from a downstream cleaning or floor-process cause.

INVESTIGATION B — FOREIGN-OBJECT CONTAMINATION

Scope	**Separate, smaller return population**
Method	**Teardown and comparative inspection against conforming parts**

A separate contamination mechanism from Investigation A. It required its own teardown protocol and comparison baseline; the corrosion investigation's isolation variables did not transfer.

WHAT I TOOK FROM IT

Two investigations, not one. Treating these as a single "contamination" bucket would have averaged two different mechanisms into one wrong answer. Splitting them is what let each get isolated correctly.

Comparative inspection needs a real baseline. Conforming parts had to be inspected alongside the returns to make "different" mean anything.

Mitigation validation is a process. Confirming a fix takes sustained data, not a single passing result.

Packaging Redesign & Machine-Vision Inspection

Mechanical & Test Engineering Co-op — Parker Hannifin, Precision Fluidics · Hollis, NH · May 2026 – Present

PACKAGING REDESIGN — TRANSIT DAMAGE ON A LOW-VOLUME COMPONENT

Trigger	**Field-reported transit damage**
Part profile	**Low-volume, high-value**
Constraint	**Fixed cost ceiling set by unit economics**

The binding constraint here was economic, not technical. A packaging fix that eliminated damage but cost more than the value it protected would never have shipped, so the design had to converge on something effective enough to stop the transit failure mode while staying inside the cost ceiling.

MACHINE-VISION INSPECTION SYSTEM — IN DEVELOPMENT

Goal	**Flag defects on a production cell before shipment**
Stack	**Python, OpenCV**
Status	**In development**

This extends the computer-vision work from my pan/tilt tracking system into a production application. A production inspection system has to run unattended against real part variation, which is a harder detection problem than tracking a fiducial marker under controlled lighting.

Printed Fixturing — Load-Frame Nest, Petri Tray & Purge Manifold

Mechanical & Test Engineering Co-op — Parker Hannifin, Precision Fluidics · Hollis, NH · May 2026 – Present

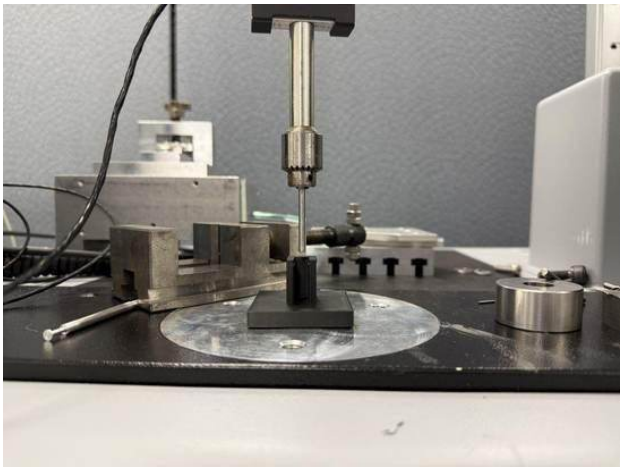
OVERVIEW

Context: Three printed fixtures supporting the same product family. One holds a molded component for mechanical characterization in a load frame, one holds petri dishes on the line, and one purges parts dry after cleaning.

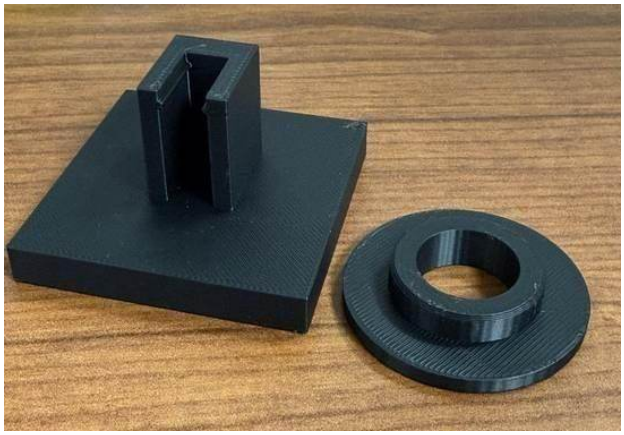
My Role: Feature and mechanism design, CAD, print iteration, and functional checkout.

LOAD-FRAME NEST

The nest exists so the part loads the same way every time. Hand-positioning a component under a probe introduces run-to-run variation in the same measurement the test is trying to resolve, so the part is constrained laterally by the slot and carried on a base wide enough to sit flat on the platen without rocking. The sealing face stays open so the probe contacts the part and nothing else. The version shown is a printed prototype; the fixture is to be rebuilt in PETG.



Nest in the load frame. Printed nest locating a molded component under the probe during sealing-force characterization. The slot constrains the part laterally while leaving the sealing face open.

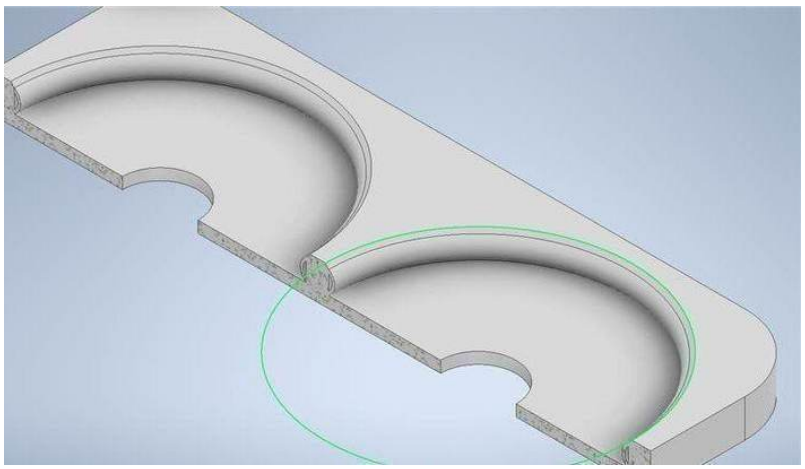


Nest and locating ring, as printed. The slotted nest sits on a wide base that registers flat to the platen. The ring is a separate locating feature printed alongside it.

COMPLIANT PETRI TRAY

A dish sitting in a plain recess is held by clearance alone, and that clearance has to be loose enough to accept dish-to-dish variation, which means the dish moves. A clamp or a spring fixes that by adding a part and a loading step. The compliant wall does it with neither: the dish deflects the wall on the way in, and the wall holds it.

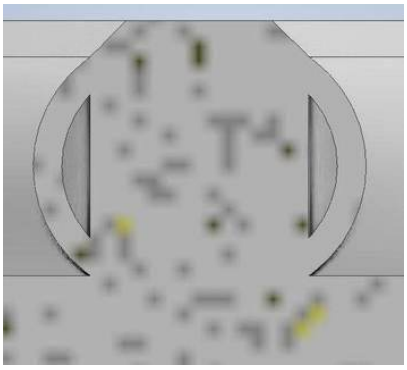
The retention wall arches over a void, with narrow slits either side acting as the hinge, so the material between them rotates rather than the wall bending as a simple cantilever. The void sets how far the wall can travel inward; the hinge sets where it gives.



Two-pocket module. Each pocket carries a compliant retention wall around the recess. Modules butt together so a tray is built from repeated units rather than drawn at a fixed length.



Section through the tray. The void behind the wall, and the section thinning toward the crown.



Flexure detail. The two slits and the neck between them, which is the material that rotates when a dish pushes on the wall.

FIRST PRINT



First print. The retention wall did not resolve at the thickness it was drawn at.

The first print did not produce the feature. The retention wall was drawn thinner than the printer can resolve, so the pockets came out as plain recesses and dishes sat loose in them. Thickening the neck to a section the process can actually build is what recovers the feature — the compliance then comes from the arch and the slit hinge rather than from a paper-thin wall.

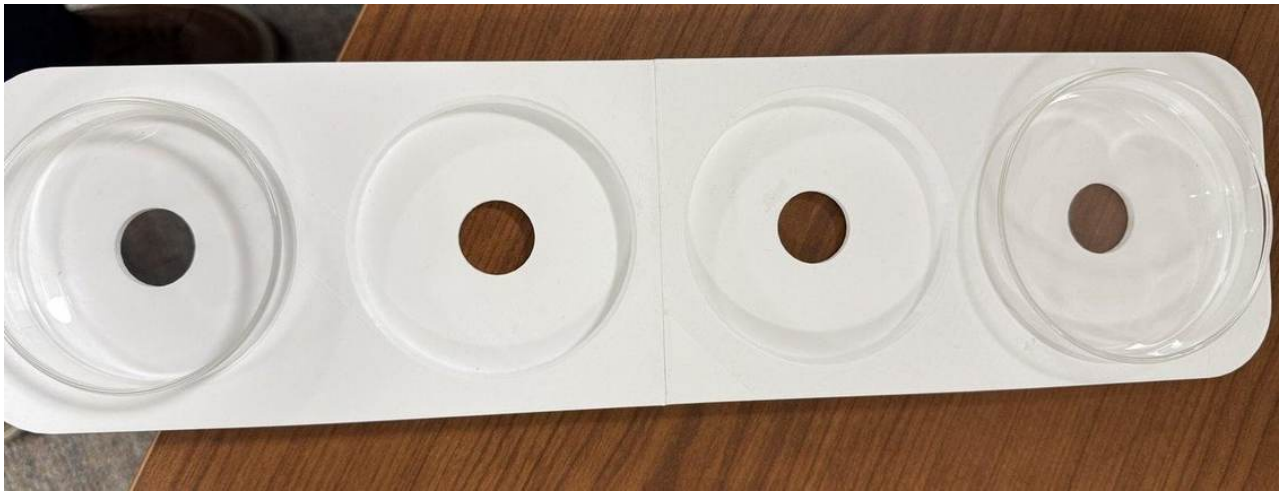
SECOND PRINT



Second print, PLA. Neck thickened to a section the slicer can build. The slit is open around each pocket and the arch behind it is present in the part.

Thickening the neck moved the wall above what the printer can resolve, and the feature came out. The slit is open around each pocket instead of closing into solid material, and the arch behind it built rather than bridging over. This build answered whether the geometry prints. It could not answer whether the wall holds, because PLA is stiffer than the design assumes and creeps under sustained deflection.

PETG BUILD



PETG build, dishes seated. Two modules snapped together, with dishes held in the outer pockets. The seam between modules is visible at centre; the joint registers the pockets of one module to the next.

The compliant wall holds the dish. A dish pushes the wall aside going in and the wall closes behind it, with no clamp, no spring, and no separate loading step. Printing it in PETG was the change that mattered: the geometry was already right after the second print, but PLA could not hold a sustained deflection long enough for the retention to mean anything.

The tray is longer than the build plate, so it prints as modules that snap together rather than as one part. Each module carries its own pockets and the snap joints register one module to the next, so tray length is set by how many modules you print instead of by the size of the printer.

DESIGN REASONING

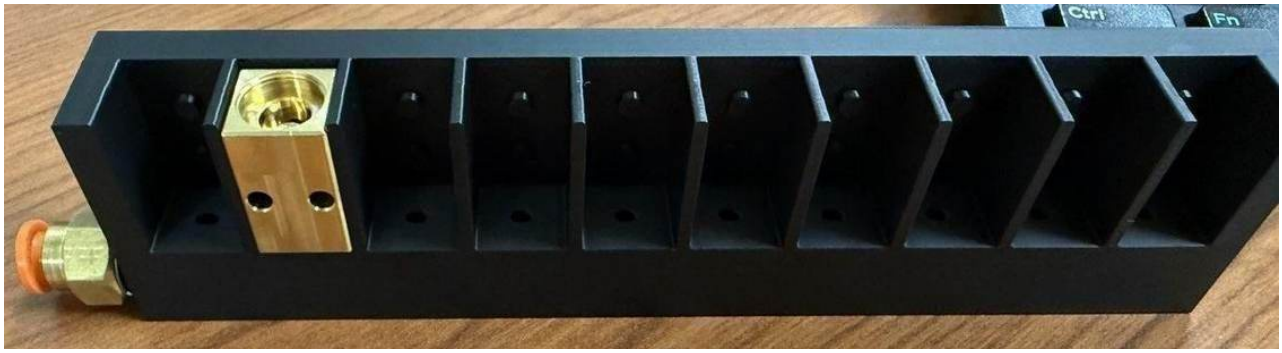
The thinnest section sits at the crown of the arch, directly over the void and between the two slit tips. That is where stress concentrates and where a failure would start, and on an FDM part it is loaded across layers rather than along them. Three builds exist: the first, where the wall never resolved; the second, where it resolved in PLA but could not be trusted to hold; and the PETG build, where it does. Each one answered exactly one question.

WHAT I TOOK FROM IT

- **A dimension is only real if the process can hold it.** The wall was drawn at a thickness no FDM nozzle can lay down, so the feature existed in CAD and nowhere else.
- **Compliance is geometry, not thinness.** Arch height and the slit hinge set the deflection. Chasing it with a thinner wall is what pushed the feature below the process floor in the first place.
- **Material is part of the design, not a substitution at the end.** The same geometry that is unreliable in PLA works in PETG, because the wall has to hold a deflection rather than survive one.
- **Print size is a design constraint.** Splitting the tray into snap-fit modules made its length independent of the build plate.

PURGE MANIFOLD

Parts coming off the cleaning step have to be dry before the next operation. Blowing them off by hand gives a different result per part and per operator, and residual moisture inside a valve does not stay where it is. It carries into whatever gets measured next.



Purge manifold. 10 stations on a common plenum, fed from a single push-to-connect inlet, with one valve body loaded. Each station locates a part over a port in the plenum floor.

One connection purges the whole batch, which turns an operator-dependent step into a single operation with the same nominal flow at every station. It is the same argument as the load-frame nest: the fixture exists to take the operator out of the variable.

Two things govern whether that claim actually holds, and both are design constraints rather than afterthoughts. Flow does not split evenly across 10 ports unless the plenum cross-section is large relative to the combined port area, so without that margin the far station dries more slowly than the near one. And an empty station is a short circuit that starves the loaded ones, so the rack either runs full or runs with plugs.

> **DESIGN REASONING**
>
> A manifold is only as good as its flow distribution. The station count is the easy part; the part that decides whether the fixture does what it claims is whether every station sees the same purge, which is a plenum sizing problem rather than a layout problem. Sizing the plenum generously relative to the ports is what buys uniformity, and it costs nothing in a printed part.

Refreshable Braille Tactile Display

Hardware Product Lab — Forge, Northeastern University · Jan 2026 – Jun 2026

OVERVIEW

Challenge: Build a refreshable Braille cell that is low-cost, survives thousands of actuation cycles, and is small enough to tile into a line of text. The governing constraint is dimensional: a six-dot cell spans 6.25 mm, and any motor worth using is wider than the cell it drives.

My Role: Mechanism architecture and encoding scheme, CAD, full drawing package, clearance and tolerance-stack iteration, prototype fabrication, and cycle testing.

DESIGN PROGRESSION

The cell went through three drive architectures. Each change was driven by a constraint the previous one could not meet.

1 — Lead screw	A lead screw integrated directly into the key system, chosen to minimize packaging by putting the drive inside the cell envelope. Dropped on time and cost before it was built.
2 — Direct rack & pinion	One motor and pinion driving a single rack beneath the pin block, lifting six dome pins off cam profiles. This is the build that was drawn, fabricated, and cycle-tested.
3 — Bevel geartrain	Two racks per cell driven through a bevel geartrain from outboard motors, with each rack encoding all eight states of its three pins. The architecture the project ended on.

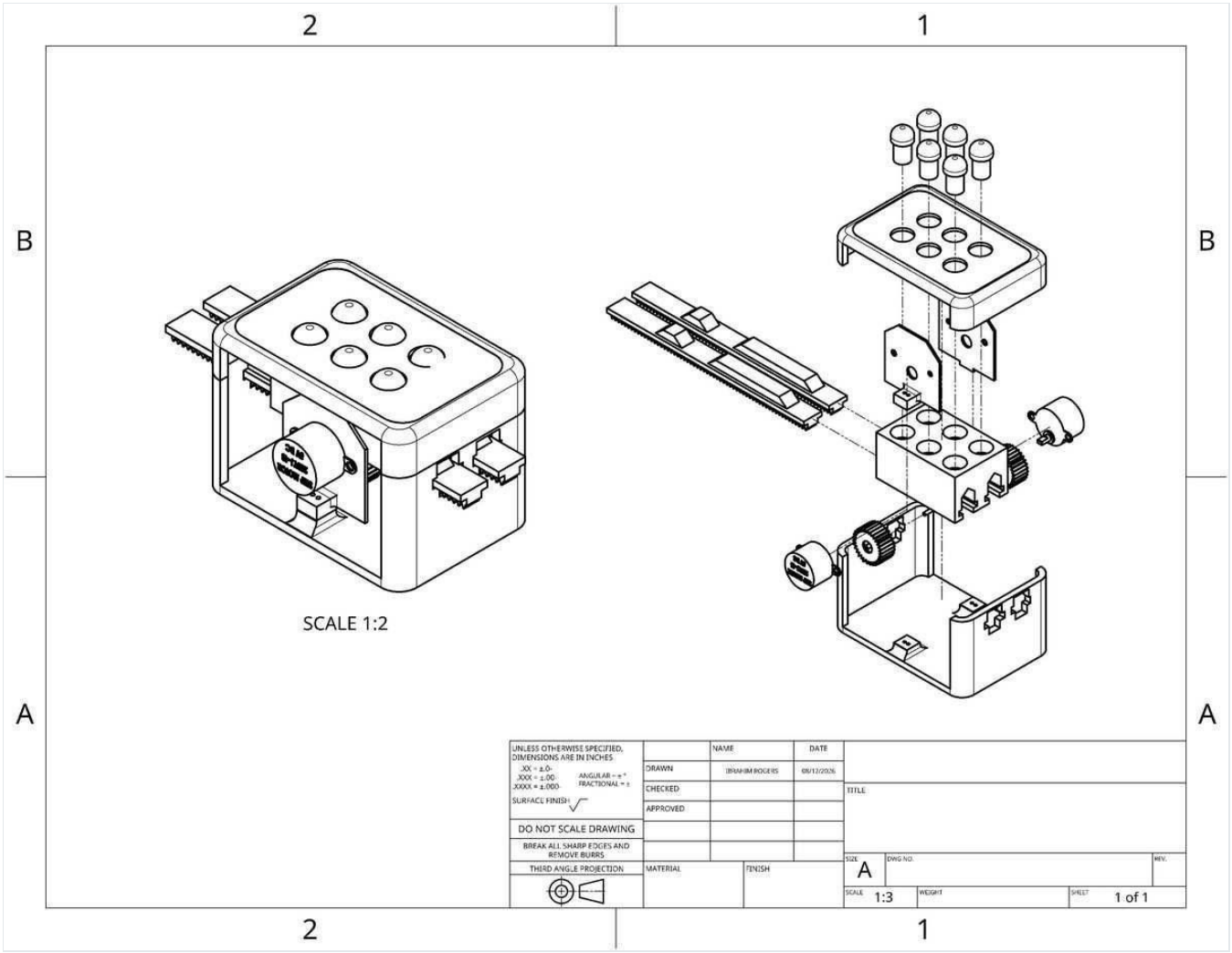
The pivot from the lead screw was a schedule and budget decision rather than a technical one, and it cost packaging density. The move from direct rack and pinion to the bevel geartrain was the opposite: it was made specifically to solve tiling, which the direct-drive layout could not do at reading pitch.

The pages that follow run in that order — the direct-drive build and its drawing package first, then the bevel geartrain architecture and the at-size hardware.

ARCHITECTURE 2 — DIRECT RACK & PINION

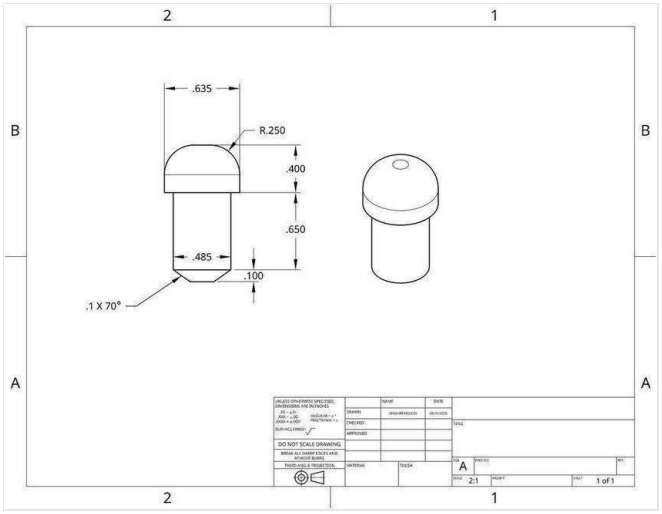
After the lead-screw concept was dropped, the cell was built as a direct-drive mechanism at enlarged scale: one motor and pinion advancing a single rack beneath the pin block, with each dot carried on its own dome pin. This is the version that was released as a drawing package, fabricated, and cycled. The drawings, tolerance work, and failure analysis on the next three pages all come from it.

FULL ASSEMBLY — EXPLODED & ISOMETRIC

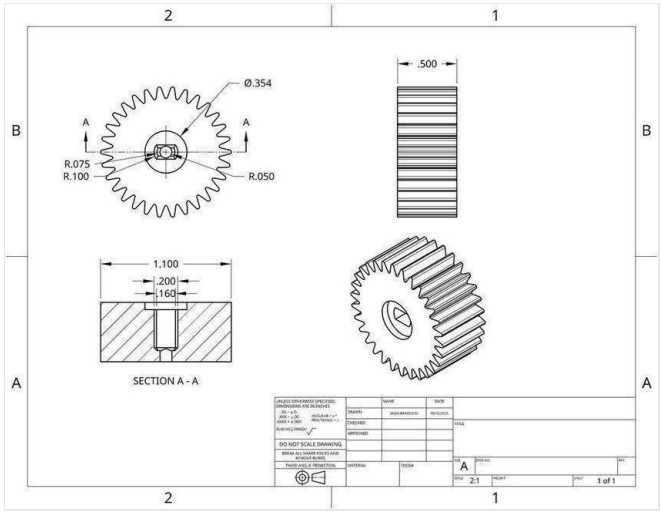


Top-level assembly drawing. Left: assembled enclosure (Scale 1:2). Right: exploded view (Scale 1:3) showing the six dome pins, letter-housing insert, guide bracket, pin-guide block, drive motor and pinion, rack, and enclosure top/bottom. Self-authored release drawing, dimensioned in inches per standard third-angle projection.

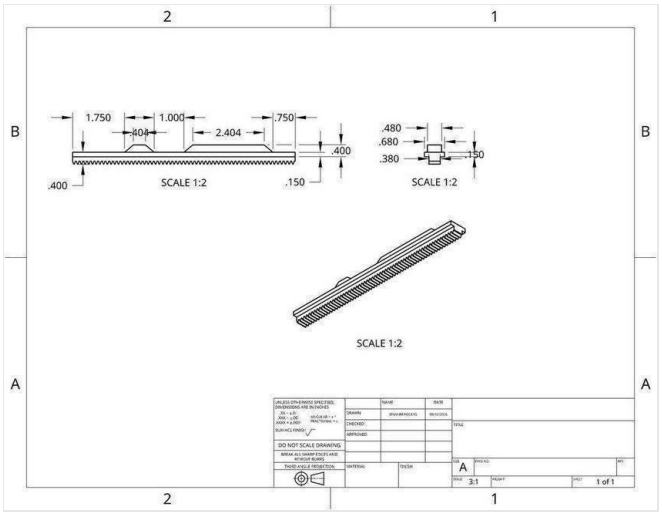
ACTUATION TRAIN — PIN, GEAR, RACK, DOWEL



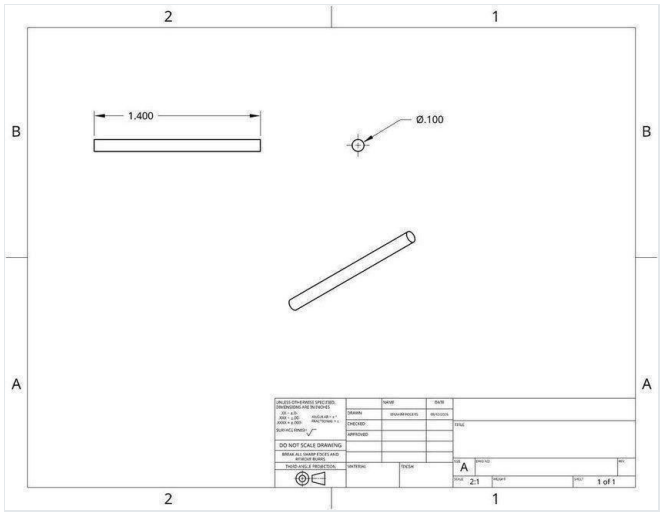
Dome pin. Ø.485 in shaft with a Ø.635 in domed head (R.250 radius) and a .1 × 70° lead-in chamfer at the tip. The shaft rides in the Ø.500 in bore of the pin-guide block, giving .015 in diametral clearance across the actuation stroke.



Drive pinion. Ø1.100 in spur gear with a Ø.354 in bore and a keyed drive slot (.200/.160 in) for the motor shaft. Root and fillet radii (R.050–R.100) are sized for the FDM-printed tooth profile rather than a cut-gear assumption.

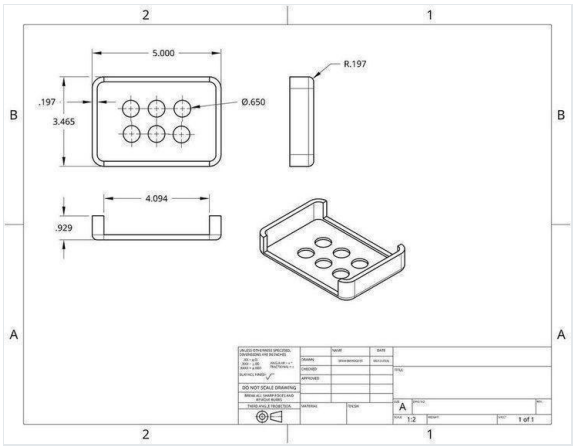


Stepped rack. Linear rack with two lead-in cam profiles at 1.000 in pitch (Scale 1:2), plus a keyed drive tab (.480/.680/.380 in, Scale 1:2) that engages the pin it lifts. This is the toleranced release version of the stepped-rack concept modeled earlier in Onshape.

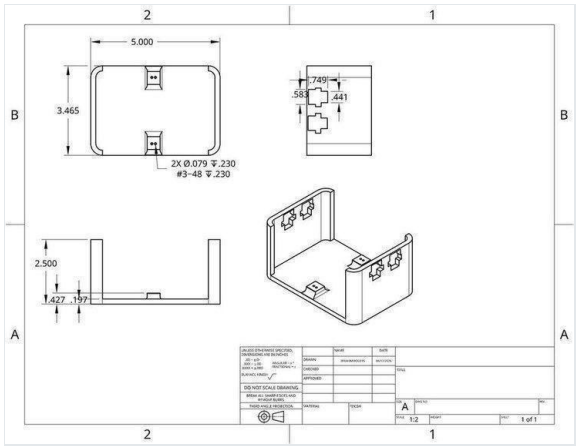


Guide dowel. Ø.100 in × 1.400 in ground pin. Locates the pin-guide block against the enclosure and keeps rack travel aligned under side load.

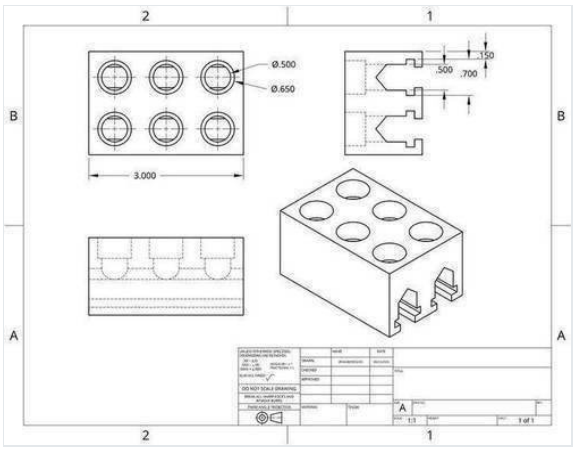
ENCLOSURE & GUIDE STRUCTURE



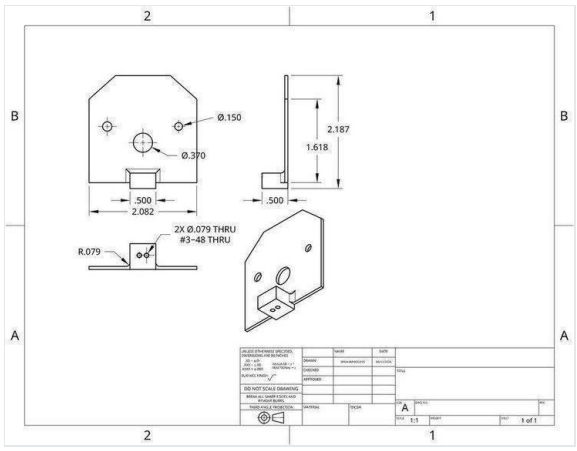
Enclosure top cover. 5.000 × 3.465 in with six Ø.650 in access holes for the dome pins and R.197 in corner fillets. Mounting flange at .929/4.094 in locates the cover to the bottom chassis.



Enclosure bottom chassis. Matching 5.000 × 3.465 × 2.500 in shell with keyed mounting tabs (2× Ø.079 in thru, #3-48 thru) for the pin-guide block and motor mount.



Pin-guide block. 3.000 in wide manifold with six bores (Ø.500 in through, Ø.650 in counterbore) carrying the dome pins, and snap-fit tabs (.150/.500/.700 in) that key it to the enclosure.



Motor-side bracket. L-bracket locating the drive motor relative to the pin-guide block, with Ø.370 in motor-shaft clearance and two Ø.150 in mounting holes on a .500 in tab.

These drawings belong to the enlarged six-pin build. Its clearance strategy does not carry over directly to the at-size cell, which uses printed features far finer than anything dimensioned here.

KEY PROBLEM — TOLERANCE STACK-UP UNDER CYCLIC LOADING

Early prototypes jammed when actuated under side load. Failure tracking isolated two compounding causes:

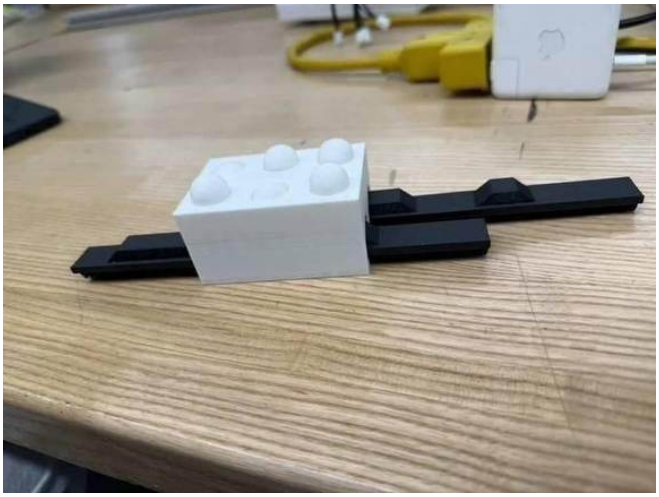
- Guide-channel clearance too tight at 0.1 mm nominal, leaving insufficient margin once print tolerance and side deflection stacked.
- Dimensional creep in PLA after sustained cycling, drifting clearances further out of spec.

Fix: opened guide clearance to 0.15 mm, added lead-in chamfers, and moved to PETG. **Result:** jam rate fell from roughly 15% to under 2% across 2,000+ test cycles. The released drawing package on the preceding pages reflects this fix directly, including the Ø.500 in pin bore sized to hold that clearance under load.

WHAT I TOOK FROM IT

- **Clearance is not just a manufacturing number.** It has to absorb print tolerance and accumulated wear, not only nominal fit.
- **Material choice is a lifecycle decision.** PLA was fine for form checks and wrong for cyclic loading, and that only showed up in testing.
- **Track failures, don't guess.** Logging jam events by condition pointed at side-load binding specifically, which made the fix targeted.
- **Design for assembly.** Consolidating cam and guide features cut part count from 12 to 8 and simplified the assembly sequence.
- **Cross-domain constraint.** Worked with the electrical side to keep actuation force inside servo torque limits while still meeting tactile force requirements.

PHYSICAL PROTOTYPE



Assembled prototype — side view. White housing carries the pin actuation mechanism and rides on the black guide rails, which provide lateral constraint. Shows rail engagement and the travel available to the rack.



Front three-quarter view. Shows the six-dot face and the housing-to-rail interface where side-load binding originated. Lead-in chamfers on the rails were added at this interface to ease pin entry.

This PLA/PETG prototype predates the released drawing package on the preceding pages and is the build that surfaced the side-load binding described there.

ARCHITECTURE 3 — BEVEL GEARTRAIN & POSITIONAL ENCODING

Direct drive put the motor in line with the cell, which set a minimum pitch far wider than a reader expects and made tiling impossible. The final architecture moves the motors outboard and drives each cell through a bevel geartrain, then uses the two resulting racks to encode the whole character instead of lifting dots one at a time.

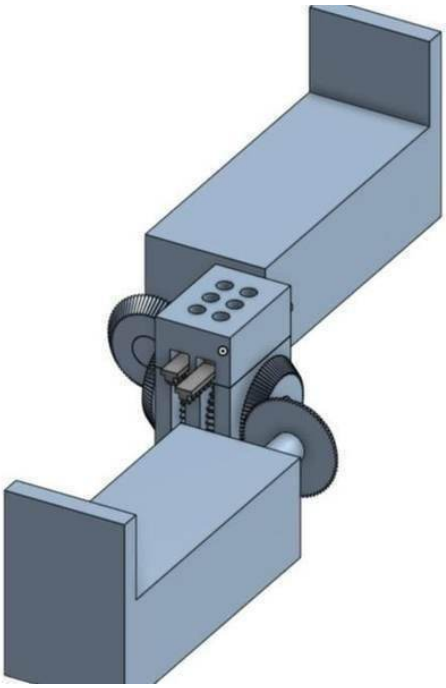
POSITIONAL ENCODING — 64 CHARACTERS, TWO MOTORS

The obvious architecture gives every dot its own actuator, six per cell. Instead, each cell carries two racks. Each rack is a stepped cam that encodes all eight raised/lowered combinations for its own set of three pins, and the motor drives the rack to whichever position the character calls for.

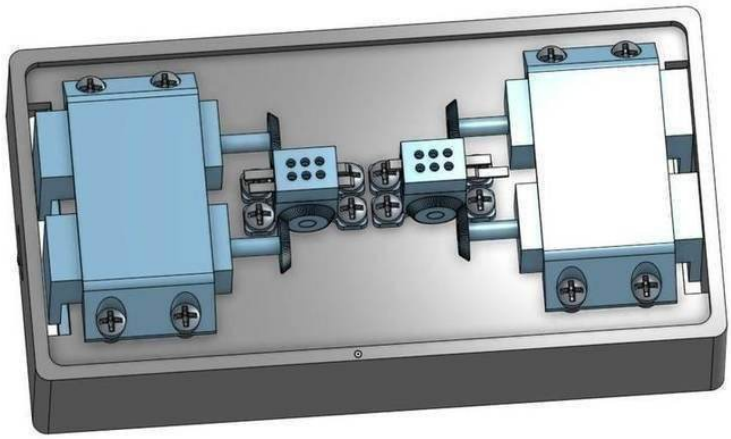
Rack position	**1 2 3 4 5 6 7 8**
Pins raised	**— a b c ab ac bc abc**

Rack A carries dots 1–2–3, rack B carries dots 4–5–6. Eight positions against eight positions is 64 states, which is exactly the six-dot Braille set including the blank cell. Actuator count per cell drops from six to two.

CELL ARCHITECTURE & TILING



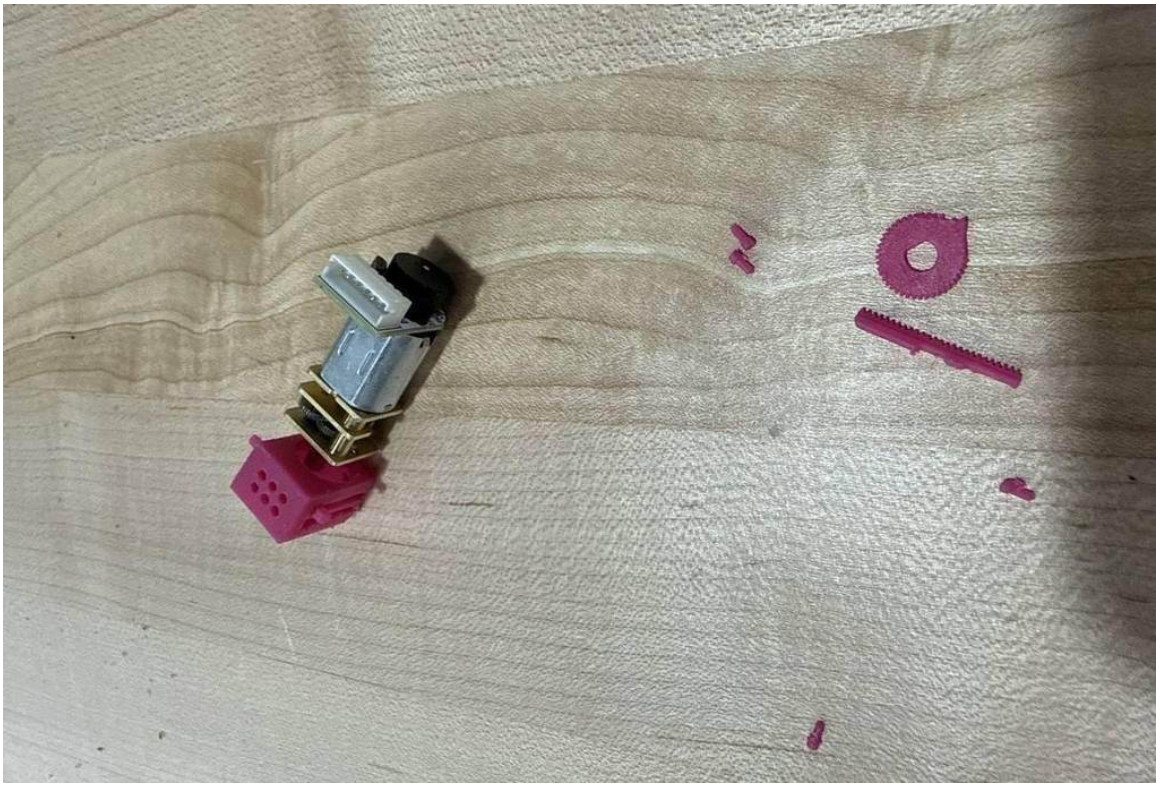
Single cell. The two racks enter from opposite faces, each driven by its own bevel gear. Rack channels extend away from the cell in opposite directions, which keeps the gearing and motors clear of the six-dot footprint.



Two-cell assembly. Cells sit adjacent at reading pitch with motors outboard and racks driving inward. Each cell is independently addressed, so a character can be rewritten without disturbing its neighbor.

Motors outboard and racks entering from opposite faces is what makes the cell tileable. Putting the actuator behind the dots would set a minimum cell pitch far wider than the 6.25 mm a reader expects, so the drive train had to leave the footprint entirely.

AT-SIZE HARDWARE



Production-scale cell, printed parts, and drive motor. The cell body sits on the output shaft of the gearmotor that drives it, with a rack, bevel gear, and dome pins alongside. Everything is shown at true relative scale, which is the point of the photograph: the drive hardware is larger than the cell it actuates.

WHY THIS WAS THE HARD PART

Actuating six dots is not difficult on its own. Actuating them inside 6.25 mm, repeatably, with hardware that can sit next to copies of itself, is the entire problem. The encoding scheme and the outboard drive layout are both answers to that constraint, not to the actuation.

Vision-Guided Pan/Tilt Tracking System

Independent Project · Dec 2025 – Present

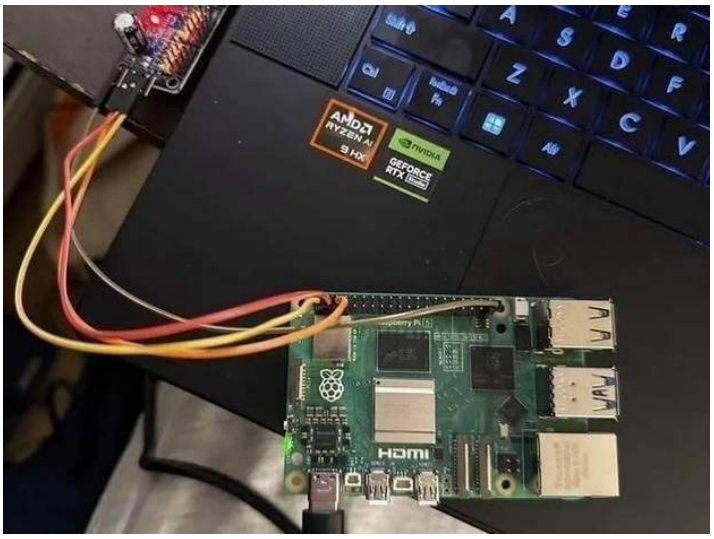
OVERVIEW

Goal: Build a closed-loop target tracking platform — mechanical gimbal, electronics, and control software — that holds a marker centered in frame and recovers on its own when the target is lost.

System: Raspberry Pi 5, camera module, PCA9685 servo driver, dual-servo pan/tilt gimbal, OpenCV ArUco detection, Python control logic, Flask live video stream.



Pan/tilt gimbal. Custom two-axis 3D-printed housing with direct-drive servo coupling. No gear train, which removes backlash from the positioning path.



Electronics integration. Raspberry Pi 5 driving the PCA9685 servo controller over I²C, with servo power on a separate rail from compute power to keep actuation transients off the Pi.

MECHANICAL DESIGN DECISIONS

- **Backlash:** direct-drive servo coupling with a press-fit interface, eliminating gear-train slop from the control loop.
- **Center of mass:** camera placement chosen to reduce servo torque load and improve response during fast target movement.
- **Rigidity:** gimbal frame reinforced to limit flex under rapid acceleration so the camera stays aligned during dynamic tracking.
- **Cable routing:** integrated to avoid wire interference across the full pan/tilt range.

INTEGRATION DEBUG — CAMERA NOT ENUMERATING

PROBLEM

No frame acquisition and no video feed. From the outside the whole system looked dead, with no indication whether the fault was mechanical, electrical, or software.

PROCESS

- Used `raspistill` to separate camera hardware faults from software faults.
- Verified I²C device enumeration to confirm the Pi was seeing devices on the bus.
- Found improper CSI ribbon seating compounded by outdated camera firmware — two independent faults presenting as one symptom.
- Updated firmware and added error handling in Python for graceful frame-drop recovery.

LESSON

Integration bugs concentrate at interface boundaries. Isolating with diagnostic tools beat guessing at the failure location, especially with two faults stacked.

CONTROL IMPLEMENTATION

- PID loop on pan and tilt axes to smooth servo motion and suppress oscillation around the target; gains tuned empirically on hardware.
- Automated re-acquisition via spiral search when the marker leaves frame or is occluded.
- Error handling for dropped frames, I^2C communication failures, and servo limit violations.

WHAT I TOOK FROM IT

- **Stage the integration.** Camera alone, then servos alone, then detection, then full loop. Bringing it up in stages saved hours over debugging everything at once.
- **Textbook PID isn't enough.** Real servo dynamics, friction, and loop latency meant the gains had to come from testing.
- **Cross-domain debugging is the actual skill.** Mechanical slop, power distribution, and software timing all produced similar-looking tracking errors.

Gamma Shielding Test System

Co-Founder & COO — Melagen Labs, VC-backed aerospace materials startup · Jan 2024 – Sep 2024

OVERVIEW

Challenge: Validate the shielding effectiveness of a melanin–HDPE composite against an aluminum reference using gamma radiation, with limited lab equipment and a startup budget.

My Role: Co-designed the test fixture, developed the experimental protocol for repeatable measurement, ran the testing, and performed the statistical analysis.

Key Result: Attenuation comparable to aluminum at roughly one-third the density, supported by ANOVA.

TEST SYSTEM & SETUP



Complete test system. Fixed-geometry fixture holds a repeatable Cs-137 source-to-detector distance. A modular sample holder allows quick swaps between composite thicknesses without disturbing the measurement geometry.

DESIGN REQUIREMENTS

- **Repeatability:** fixed source-to-detector positioning to minimize geometric variation between runs, the dominant error source in this measurement.
- **Modularity:** quick-change sample mounting to test multiple thicknesses and compositions efficiently.
- **Safety:** Cs-137 source containment and shielding per lab safety protocol.
- **Low-cost fabrication:** built from lab-available materials within startup budget.

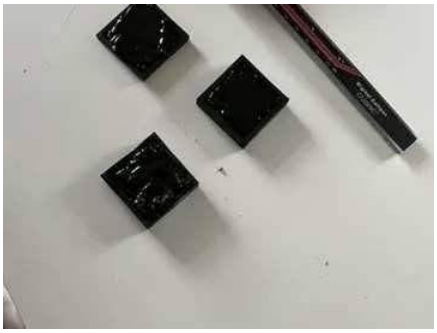
SAMPLE PREPARATION & MEASUREMENT



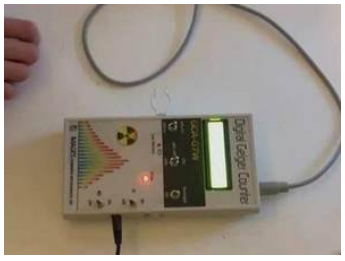
Material preparation. Melanin powder measured to a controlled mass so concentration stayed consistent batch to batch.



Sample fabrication. Composite cast in aluminum pans for consistent thickness, under controlled material-handling protocol.



Test specimens. Composite plates at varying thickness, measured and tested against aluminum references.



Radiation measurement. Calibrated Geiger counter at fixed distance from the shielded source, logging count rate for each shielding condition against an aluminum baseline.

ENGINEERING PROCESS

- **Fixture design for repeatability:** hand-positioning produced unacceptable run-to-run measurement scatter. Adding fixed mechanical alignment features cut that variation substantially and was the single biggest improvement to data quality.
- **Rapid iteration under constraints:** when the first fixture drifted after repeated sample changes, redesigned the mounting within days using available shop tools.
- **Statistical validation:** ANOVA across material types and thicknesses, with regression modeling of shielding effectiveness against melanin concentration.
- **Uncertainty analysis:** documented contributions from source activity calibration, detector efficiency, and geometric positioning so the results held up to outside review.
- **Technical communication:** translated the data into engineering presentations used in fundraising.

WHAT I TOOK FROM IT

- **Fixture design drives data quality.** In experimental work, repeatability is the whole game. Small mechanical improvements to alignment had far more impact on confidence in the result than anything downstream in the analysis.
- **Constraints force decisions.** Limited budget and equipment meant choosing what actually mattered to measure well.

Technical Skills

Tools and methods I can speak to in detail and have applied on the work in this portfolio

CAD & DESIGN

Autodesk Inventor · SolidWorks · Onshape · AutoCAD
GD&T (MMC/LMC) · worst-case tolerance stack-up · interference analysis · fixture design · engineering drawings · P&ID development

FLUID & TEST SYSTEMS

Hydraulic/pneumatic architecture · pressure regulation and relief · closed-loop recirculation · pump and chiller sizing · manifold design · pressure-decay and flow-based leak test methods · test-stand architecture

PROGRAMMING & ANALYSIS

Python (OpenCV, machine-vision inspection, pandas) · C++ · MATLAB · Excel
ANOVA / regression · I^2C and PWM interfaces · Raspberry Pi · Arduino · hardware–software integration

MANUFACTURING & PROTOTYPING

FDM 3D printing (PLA, PETG) · SLA/resin printing · rapid prototyping · mechanical assembly · fixture fabrication · design for manufacturing · design for assembly

QUALITY & VALIDATION

Failure analysis · root-cause investigation · FMEA · dimensional inspection and metrology · experimental test design · nonconformance investigation · validation and test documentation

ENGINEERING COMMUNICATION

Failure-analysis reporting · design reviews · cross-functional requirements definition · vendor and supplier coordination · technical presentations to non-technical audiences

ADDITIONAL PROJECT EXPERIENCE

Assistive Sweeper Robot
Cornerstone II, London · Jan–Apr 2025

Co-designed an RC sweeper in SolidWorks, working drivetrain geometry against motor torque and ground clearance constraints. Iterated through a rapid prototyping sprint for operator usability and assembly.

Autonomous Maze Robot
Cornerstone I, Oakland · Sep–Dec 2024

Programmed sensor-based autonomous navigation logic. Diagnosed drivetrain binding and electrical contact failures through systematic root-cause analysis across a two-week test cycle.

Hydrogen Fuel Cell RC Vehicle
TKS Innovation Fellow · 2023–24

Designed a hydrogen fuel cell power system, managing power distribution and integration. Won a Google-sponsored pitch competition for the technical proposal.

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